Brian Lee

Fairfax, VA • lee.brian.j9@gmail.com • (571) 279-7134 • leebri2n.github.io

PROFESSIONAL SUMMARY

ML engineer with 2+ years building evaluation frameworks and data pipelines for AI-enabled systems, including computer vision and generative models. Specialized interest in explainable AI and uncertainty quantification for AI systems. Experienced with GPU-accelerated training, HPC environments, and Python packaging with multiple peer-reviewed IEEE publications.

EXPERIENCE

Virginia Tech National Security Institute — **Arlington, VA**
Research Data Analyst — *Aug. 2023 – Aug 2025*

- **ML testing software:** Led development of CODEX, an open-source Python tool implementing combinatorial coverage for ML data analysis. Refactored **~85%** of the previous codebase into a modular OOP design, built analytics/visualization and ML experimentation modules, and applied the tool to cross-domain experiments.
- **ML systems integration:** Implemented computer vision pipelines for MIMIK, an open-source mission engineering tool, using YOLOv8 classification models. Integrated pipelines into the simulation framework to emulate AI agent perception in command chain modeling.
- **Deep learning experimentation:** Trained, fine-tuned, and debugged deep-learning training pipelines to enable experiments on dataset bias and on the characterization of ML operating envelopes.

Virginia Tech National Security Institute — **Arlington, VA**
Research Intern — *May 2022 – Aug. 2022*

- **Data pipeline engineering:** Built an image pipeline in Python that automated scraping and preprocessing of **150k+** images from Instagram at **~120** images/min. Assembled **50+** low-resource datasets for data-efficient generative modeling experiments.
- **Generative model experiments:** Designed controlled StyleGAN3 training experiments comparing transfer learning to from scratch baselines when applied to limited datasets, achieving up to **1.5**× faster FID convergence.

SKILLS

Languages: Python; Java; C/C++; MATLAB
ML / Deep Learning: PyTorch; Tensorflow; NumPy; Pandas; scikit-learn; StyleGAN3; YOLO
ML Areas: Data assurance: robustness, fairness, and representation analysis; Low-resource training; Computer vision
HPC / Parallel: CUDA; Batch scripting; MPI; OpenMP; SLURM

PROJECTS

CODEX (Coverage of Data Explorer) | Open-source · github.com/vtnsi/codex — *Aug. 2023 - May 2025*

- Built an output module that generates metrics reports, coverage and skew plots, and coverage-to-performance figures using Matplotlib and Seaborn.
- Integrated training pipelines for select models (incl. YOLO), enabling data-to-performance testing capabilities.
- Improved end-user usability by revamping CLI and adding Python API class-based options for running coverage analyses.

Comparing Representations by Protein Sequencing Models (Capstone) — *Aug. 2022 - Dec. 2022*

- Applied Gaussian Mixture Model clustering (scikit-learn) across **500k+** protein sequences to compare representation similarity between sequencing models.

PUBLICATIONS

- Mhatre, S., **Lee, B**., et al. "Coverage Metrics for Detecting Bias in Facial Recognition Datasets," in *2025 IEEE Conf. on AI, 2025*. Presented May 2025.
- Lanus, E., **Lee, B**., et al. "CODEX: Testing Machine Learning with the Coverage of Data Explorer Tool," in *2025 IEEE conf on AI Testing*, 2025. Presented August 2025. Demos presented at NIST's CT4AIES 2024 and Dataworks 2025.

EDUCATION

Virginia Tech — **Blacksburg, Virginia**
B.S. in Computational Modeling and Data Analytics — *May 2023*
Minors in Computer Science and Mathematics